

Equations of Motion (EOM) Derivation

Mathematical Modeling of a Hexapod Robot

By: Prachi Gopani [H021]

Hrishikesh Gudekar [H022]

Daksh Hariya [H023]

1. Introduction

Hexapod robots are bio-inspired locomotion systems that replicate the walking mechanisms of insects and crustaceans. These robots are capable of stable locomotion because their six-leg configuration allows multiple legs to maintain contact with the ground at all times.

Each leg of a hexapod robot typically consists of three rotational joints: the coxa, femur, and tibia. These joints allow the robot to move its legs in different directions and generate the walking motion required for locomotion.

To analyze the motion and dynamics of the robot, it is necessary to develop a mathematical model describing the relationship between joint torques, forces, and resulting motion. In this work, the equations of motion (EOM) for a hexapod leg are derived using the Lagrangian method, which is widely used in robotics and dynamic system modeling.

The derived equations will later be useful for simulation, gait planning, stability analysis, and control system design.

2. System Description

Each leg of the hexapod robot is modeled as a three-link serial manipulator.

The joints are:

Joint	Link	Function
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Coxa	Link 1	rotates the leg sideways
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Femur	Link 2	lifts the leg up and down
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Tibia	Link 3	extends the leg
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The links are connected by revolute joints driven by motors.

Let

- l_1 = length of coxa link
- l_2 = length of femur link
- l_3 = length of tibia link

Joint angles:

- θ_1 = coxa joint angle
- θ_2 = femur joint angle
- θ_3 = tibia joint angle

These joints provide 3 degrees of freedom (3-DOF) for each leg.

3. Modeling Assumptions

To simplify the modeling process, the following assumptions are made:

1. The robot body is treated as a rigid body.
2. Each leg is modeled as a three-link planar manipulator.
3. The ground surface is assumed to be flat and rigid.
4. The foot does not slip during ground contact.
5. Air resistance is neglected.
6. Friction at the joints is negligible.
7. The center of mass of each link lies at the geometric center of the link.

These assumptions reduce the complexity of the dynamic equations.

4. Coordinate System and Generalized Coordinates

To describe the configuration of the robot, a coordinate system is defined.

A world coordinate frame is attached to the ground.

A body coordinate frame is attached to the center of the robot body.

The configuration of one leg is defined by the generalized coordinates:

$$q = [\theta_1, \theta_2, \theta_3]$$

where

Variable	Meaning
θ_1	coxa joint angle
θ_2	femur joint angle
θ_3	tibia joint angle

These angles fully describe the leg configuration.

5. Choice of Dynamic Modeling Method

The **Lagrangian formulation** is used to derive the equations of motion.

The Lagrangian method is preferred because:

- It uses **generalized coordinates**, which simplifies modeling of multi-link robotic systems.
- It avoids calculating forces individually for each link.
- It provides a systematic approach to deriving dynamic equations.

Therefore, the Lagrangian method is well suited for **multi-joint robotic systems such as hexapod legs**.

6. Kinematic Model of the Leg

The position of the foot relative to the hip joint is determined using forward kinematics.

Let

- l_1, l_2, l_3 be the link lengths.

The position of the foot in Cartesian coordinates is:

$$x = l_1 \cos \theta_1 + l_2 \cos (\theta_1 + \theta_2) + l_3 \cos (\theta_1 + \theta_2 + \theta_3)$$

$$y = l_1 \sin \theta_1 + l_2 \sin (\theta_1 + \theta_2) + l_3 \sin (\theta_1 + \theta_2 + \theta_3)$$

These equations describe the position of the foot based on joint angles.

7. Velocity Derivation

Velocity is obtained by differentiating the position equations with respect to time.

For the x-direction:

$$\dot{x} = -l_1 \sin \theta_1 \dot{\theta}_1 - l_2 \sin (\theta_1 + \theta_2)(\dot{\theta}_1 + \dot{\theta}_2) - l_3 \sin (\theta_1 + \theta_2 + \theta_3)(\dot{\theta}_1 + \dot{\theta}_2 + \dot{\theta}_3)$$

For the y-direction:

$$\dot{y} = l_1 \cos \theta_1 \dot{\theta}_1 + l_2 \cos (\theta_1 + \theta_2)(\dot{\theta}_1 + \dot{\theta}_2) + l_3 \cos (\theta_1 + \theta_2 + \theta_3)(\dot{\theta}_1 + \dot{\theta}_2 + \dot{\theta}_3)$$

These equations represent the velocity of the foot.

8. Acceleration Derivation

Acceleration is obtained by differentiating velocity with respect to time.

This produces second-order derivatives of the joint angles.

The acceleration of the foot is expressed as:

$$\begin{aligned}\ddot{x} &= \frac{d}{dt}(\dot{x}) \\ \ddot{y} &= \frac{d}{dt}(\dot{y})\end{aligned}$$

These equations are necessary for deriving the dynamic equations.

9. Forces Acting on the System

The following forces act on the hexapod leg.

Gravitational Force

$$F_g = mg$$

where

- m = mass of the link
 - $g = 9.81 \text{ m/s}^2$
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Ground Reaction Forces

When the foot contacts the ground, reaction forces act on the leg.

$$F = [F_x, F_y]$$

where

- F_x = horizontal ground force
 - F_y = vertical ground force
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Joint Actuator Torques

Motors at the joints apply torques.

$$\tau = [\tau_1, \tau_2, \tau_3]$$

where

Torque Joint

τ_1	coxa
τ_2	femur
τ_3	tibia

10. Kinetic Energy

The kinetic energy of the system is

$$T = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2$$

where

- m_1, m_2, m_3 = masses of links
 - v_1, v_2, v_3 = velocities of the centers of mass.
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11. Potential Energy

Potential energy due to gravity is

$$V = mgh$$

where

- h = height of the center of mass.

12. Lagrangian Formulation

The Lagrangian of the system is defined as

$$L = T - V$$

The equations of motion are derived using:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i$$

where

- q_i = generalized coordinate
- Q_i = generalized force.

13. Final Dynamic Equation of Motion

After applying Lagrange's equation for all coordinates, the dynamic model becomes

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau$$

where

Term	Meaning
$M(q)$	mass matrix
$C(q, \dot{q})$	Coriolis and centrifugal forces
$G(q)$	gravity vector
τ	actuator torque

14. State-Space Representation

Define the state vector

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$

The system can be written as

$$\dot{x} = \begin{bmatrix} \dot{q} \\ M^{-1}(q)(\tau - C(q, \dot{q})\dot{q} - G(q)) \end{bmatrix}$$

This representation is useful for **simulation and control design**.

15. Unit Consistency Check

Quantity	Unit
Angle	rad
Velocity	rad/s
Acceleration	rad/s ²
Force	N
Torque	N·m
Mass	kg

All equations are dimensionally consistent.

16. Physical Meaning of Dynamic Terms

For the equation

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau$$

M(q) represents inertia of the links.

C(q, $\dot{q}$) represents velocity-dependent forces.

G(q) represents gravitational forces.

τ represents torques applied by motors.

17. Conclusion

A dynamic model of a hexapod robot leg with three joints (coxa, femur, tibia) was developed. The kinematic equations describing foot position were derived, followed by velocity and acceleration expressions. Using the Lagrangian formulation, the equations of motion were obtained.

The final equations describe the relationship between joint torques and leg motion, providing a foundation for simulation, gait design, and control of the hexapod robot.